TUTORIAL 2 (P3)

CHAPTER 2: FURTHER ALGEBRA (PARTIAL FRACTIONS & BINOMIAL EXPANSION)

Section I – Notes and Exercises

2.1: PARTIAL FRACTIONS

Introduction to Partial Fractions

1. A **rational function** is an **algebraic fraction** such that **both the numerator and the denominator are polynomials**.
2. The algebraic fraction $\frac{3}{x+2} - \frac{2}{2x-1}$ can be expressed as a single fractional function, that is,

$$\frac{3}{x+2} - \frac{2}{2x-1} = \frac{3(2x-1) - 2(x+2)}{(x+2)(2x-1)}$$
$$= \frac{4x-7}{(x+2)(2x-1)}$$

3. The **reverse process** to **decompose** the rational function $\frac{4x-7}{(x+2)(2x-1)}$ into $\frac{3}{x+2} - \frac{2}{2x-1}$ is called '**expressing** $\boldsymbol{\frac{4x-7}{(x+2)(2x-1)}}$ **in partial fractions'**.

Proper and Improper Rational Functions

1. Consider a rational function of the form $\frac{f(x)}{g(x)}$, where $f(x)$ and $g(x)$ are polynomials in x.
2. If **degree of $\boldsymbol{f(x)}$ < degree of $\boldsymbol{g(x)}$,** then $\frac{f(x)}{g(x)}$ is called a **proper rational function**.
3. If **degree of $\boldsymbol{f(x)}$ ≥ degree of $\boldsymbol{g(x)}$,** then $\frac{f(x)}{g(x)}$ is called an **improper rational function**.

Expressing a Rational Function in Partial Fractions: The Procedure

Step 1: Determine the form of the partial fraction
Based on the type of factors in the **denominator**, there are 4 different **rules** to follow.

Rule 1: **Distinct linear factors** in the denominator
Example: $\frac{2x+3}{(x+3)(x-6)} \equiv \frac{A}{x+3} + \frac{B}{x-6}$

Rule 2: **Repeated linear factor** in the denominator
Example: $\frac{x^2}{(x-3)^3} \equiv \frac{A}{x-3} + \frac{B}{(x-3)^2} + \frac{C}{(x-3)^3}$

Rule 3: **Irreducible quadratic factor** in the denominator
Example: $\frac{3x^2+2x}{(x+2)(x^2+3)} \equiv \frac{A}{x+2} + \frac{Bx+C}{x^2+3}$

Rule 4: **Improper rational functions (Degree of numerator ≥ Degree of denominator)**
Example: $\frac{x^2+3x}{x^2-4}$
First, perform **polynomial division**. Then we have

$$\frac{x^2+3x}{x^2-4} = Quotient + \frac{Remainder}{x^2-4}$$

$\frac{Remainder}{x^2-4}$ is a proper rational function which can be expressed in partial fractions using Rule 1, Rule 2 or Rule 3. [***Question***: In this case, which of the 3 Rules should we use?]